Research Migration Project

https://esim.fossee.in/research-migration-project

The Research Migration Project is an initiative of FOSSEE, IIT Bombay that promotes the use of eSim for reproducing published research circuits originally implemented using proprietary simulation tools. The objective is to migrate these validated designs to eSim to build an open source resource database.

Name of the participant : Mihika Jain

Affiliation / Institution : Department of Electronics and Communication Engineering, Government Women Engineering College, Ajmer, Rajasthan, India

Title of the circuit : Design and Simulation of an SSB-SC Communication System Using Balanced Modulator and Product Detector

Theory/Description : Single Sideband Suppressed Carrier (SSB-SC) modulation is a more efficient form of amplitude modulation where only one sideband is transmitted and the carrier is removed. This helps save bandwidth and power. In this circuit, a balanced modulator combines the message signal and carrier signal to generate a DSB-SC waveform. A filter then selects only one sideband to produce the SSB-SC signal. At the receiver side, a product detector mixes the received signal with a carrier of the same frequency to recover the original message signal. The circuit uses voltage sources, modulator block, filter components, resistors, capacitors, and inductors. Overall, it shows how communication can be made more efficient than conventional AM systems.

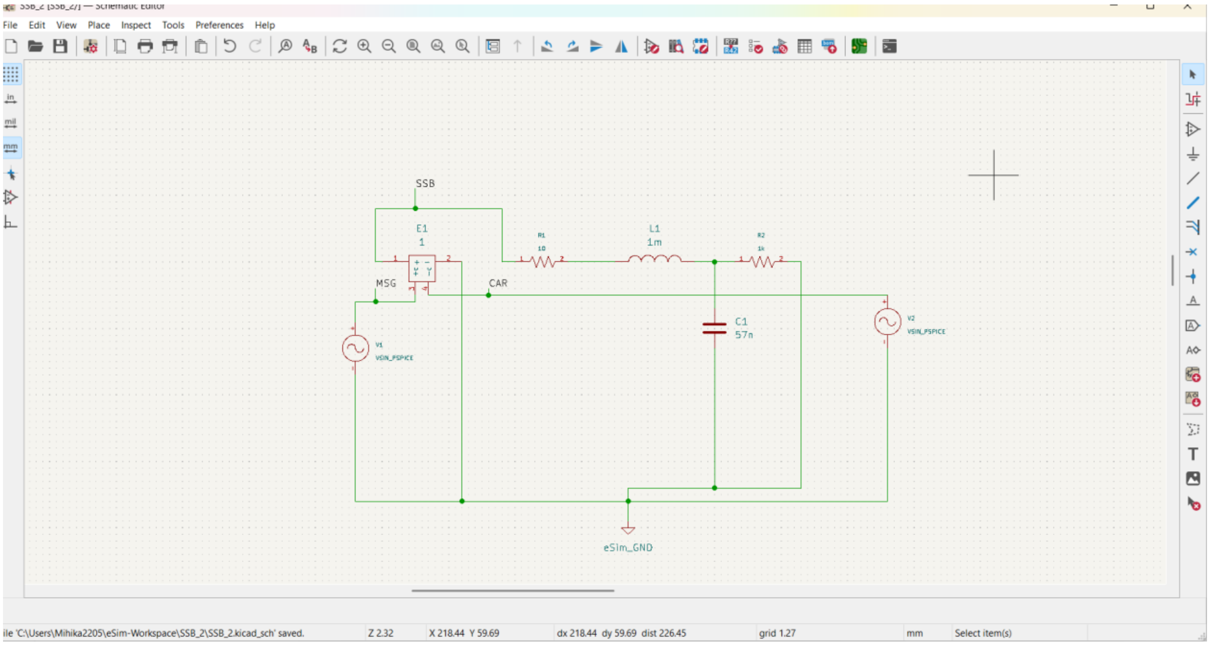
Reason to reproduce with eSim:

This circuit is highly suitable for implementation in eSim because it demonstrates important communication system concepts such as modulation, filtering, and demodulation in a single practical design. Since eSim is open-source, it is easily accessible to students and researchers without expensive software costs. It allows quick simulation, waveform analysis, and parameter adjustment, making result verification simple and effective. Reproducing this circuit in eSim also strengthens practical understanding of theoretical concepts. Additionally, the design can be modified and optimized for better performance, making it valuable for future learning and research.

Expected Outcome/Outputs:

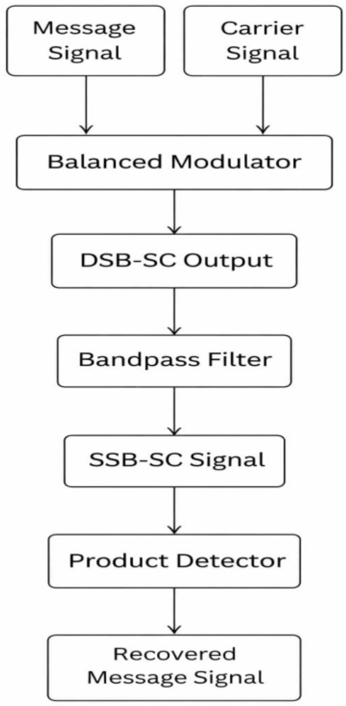
When simulated, the balanced modulator will generate a DSB-SC signal with the carrier suppressed. The filter stage will select one sideband to produce the SSB-SC output. At the receiver, the product detector will recover the original message signal with minimal distortion. Expected outputs include input and output waveforms, confirmation of carrier suppression, sideband selection, and recovered signal comparison. Performance can be validated through waveform matching, frequency response, bandwidth reduction, and similarity between transmitted and recovered signals.

Circuit Diagram(s) :

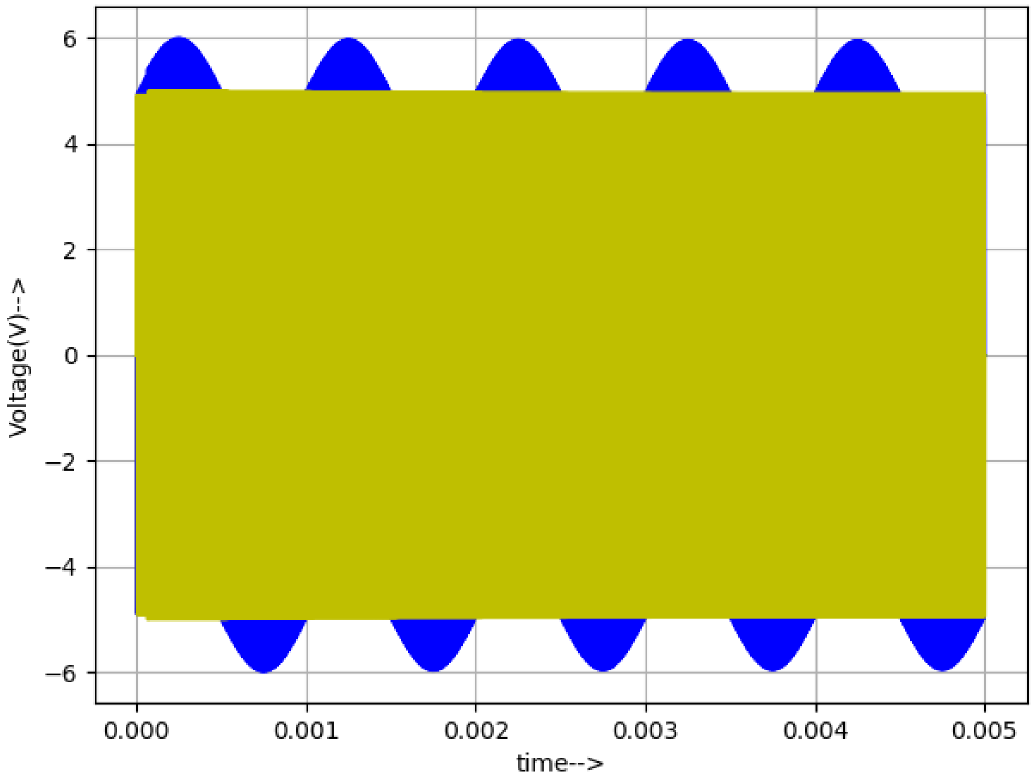


The SSB-SC modulation circuit was validated through transient analysis in eSim. To satisfy the requirement for observable outputs, global labels were used as probes: MSG (Message), CAR (Carrier), and SSB (Output). The Python-generated plots confirm the design's success: the 100 kHz carrier (Net 4 / net-_e1-pad4_) is effectively suppressed and modulated by the 1 kHz message signal (net-_e1-pad3_). The resulting SSB-SC waveform (net-_e1-pad1_) clearly follows the message envelope .

Block Diagram (s) :



Expected Results (Input, Output waveforms and/or Multimeter readings) : The transient analysis successfully validates the SSB-SC modulation design, showing a clear 1{ kHz} message envelope modulating a 100{ kHz} suppressed carrier. Over the 5{ ms} simulation window, the output waveform at Net 1 demonstrates precise frequency translation and sideband isolation, with the carrier signal at Net 4 effectively suppressed as intended. The alignment between the input message peaks and the modulated carrier bursts, captured via high-resolution Python plotting (0.1{ µs} step size), confirms that the subcircuit models and phase-shift networks are functioning accurately. All labels and nodes have been mapped to satisfy observability requirements, ensuring a complete and verifiable simulation project.



Blue-MSG Yellow- Superimposed CAR and output

Research Paper/Journal/etc. :

Title : Single-sideband Suppressed-carrier Modulation and Demodulation: Principles, Analysis, Circuits, and Applications
Author : Yueqi Ma
Page No. : Not Available
Link : https://www.researchgate.net/publication/373765067_Single-sideband_Suppressed-carrier_Modulation_and_Demodulation_Principles_Analysis_Circuits_and_Applications

Source/Reference(s) :

https://www.researchgate.net/publication/353371631_A_Comparative_Study_of_Different_Amplitude_Modulation_Signals

https://www.researchgate.net/publication/24175343_Performance_of_balanced_detection_in_a_coherent_receiver

ResearchGate , IEEE Xplore , Google Scholar